

LINKSFOUNDATION.COM



**Politecnico
di Torino**

Applied Data Science Project

L1 - Introduction



Course objective

Building an artificial intelligence solution with data science techniques and methodologies to solve real-world problems



Students' feedback

Organizzazione dell'insegnamento

Il corso fantastico, teorico al punto giusto e lascia il giusto spazio alle attività pratiche. E' organizzato molto bene, si vede che il prof. Rizzo ha passione e migliora il corso di anno in anno. Ho feedback assolutamente positivi, e ringrazio infinitamente il prof. per permettere di seguire da casa e svolgere i checkpoint da remoto (lavorando e studiando da remoto, mi sarebbe venuto molto complicato partecipare ai checkpoint in presenza e ciò mi ha agevolato moltissimo).

2023-2024

Efficacia del/della docente

great

I have had an impressive journey in this course. it gives me a relatively complete concept of the overall project process. I'm glad to attend a course like this.



Il corso ha avuto solo aspetti positivi e non mi sento dunque di suggerire miglioramenti perchè, per quel che mi riguarda, è stato completo così come è.



2022-2023

I am glad to have an opportunity to study with a company on their project professionally. Also this course provide an environment to practice how to organize a team work and present the done work properly.

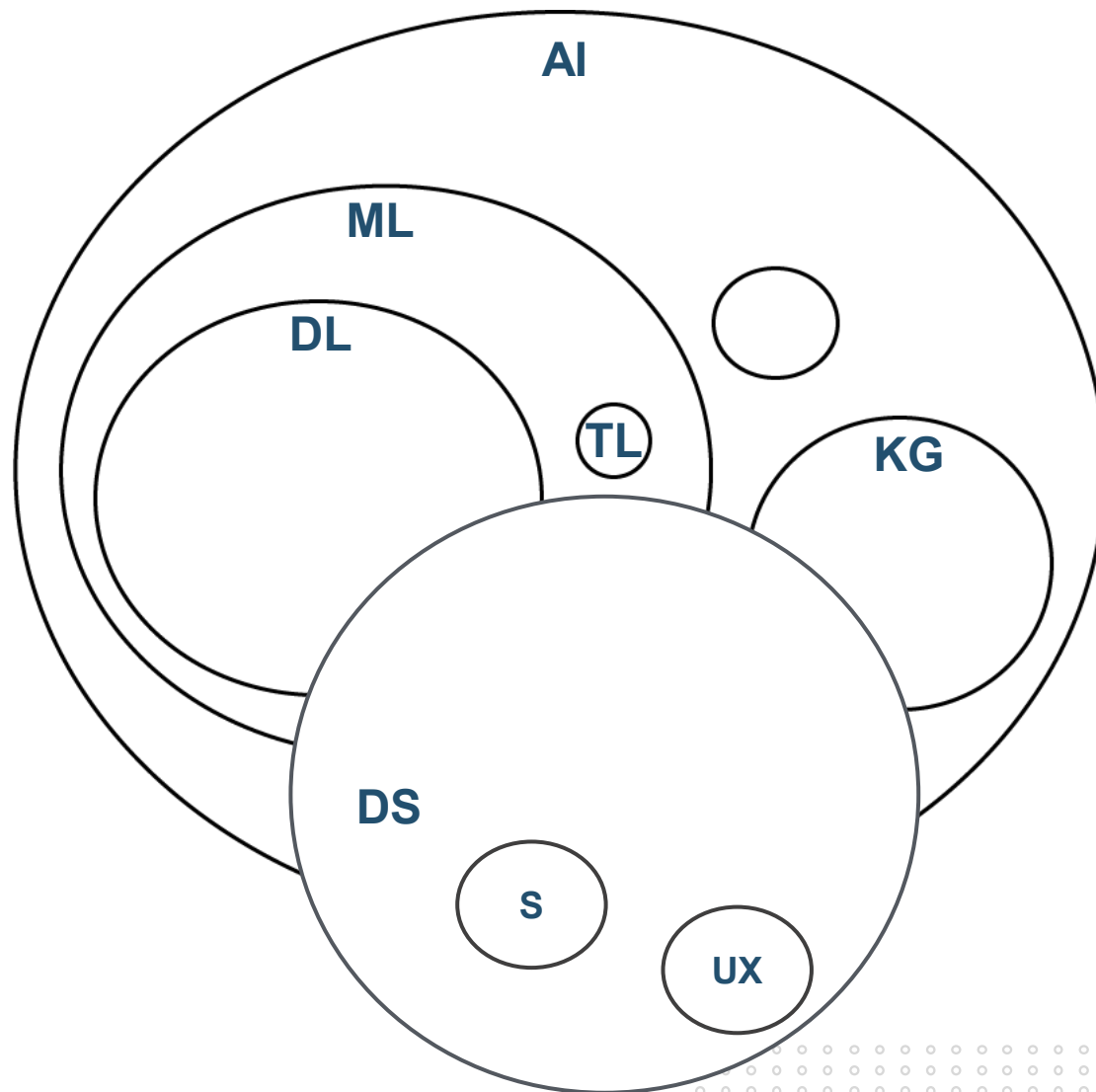


This is an extremely helpful course. I am feeling lucky to attend this course. After completing it I felt that I earned valuable experiences in the field of Data Science. I am feeling ready to professional life more than ever. Especially working with the professional project management and collaboration tools is one of the best side of this course. Honestly I can find anything to suggest, I hope this course remains in the curriculum of our program. So everyone can have the same experience.



2021-2022

Interplay



AI: Artificial Intelligence

ML: Machine Learning

DL: Deep Learning

TL: Transfer Learning

KG: Knowledge Graph

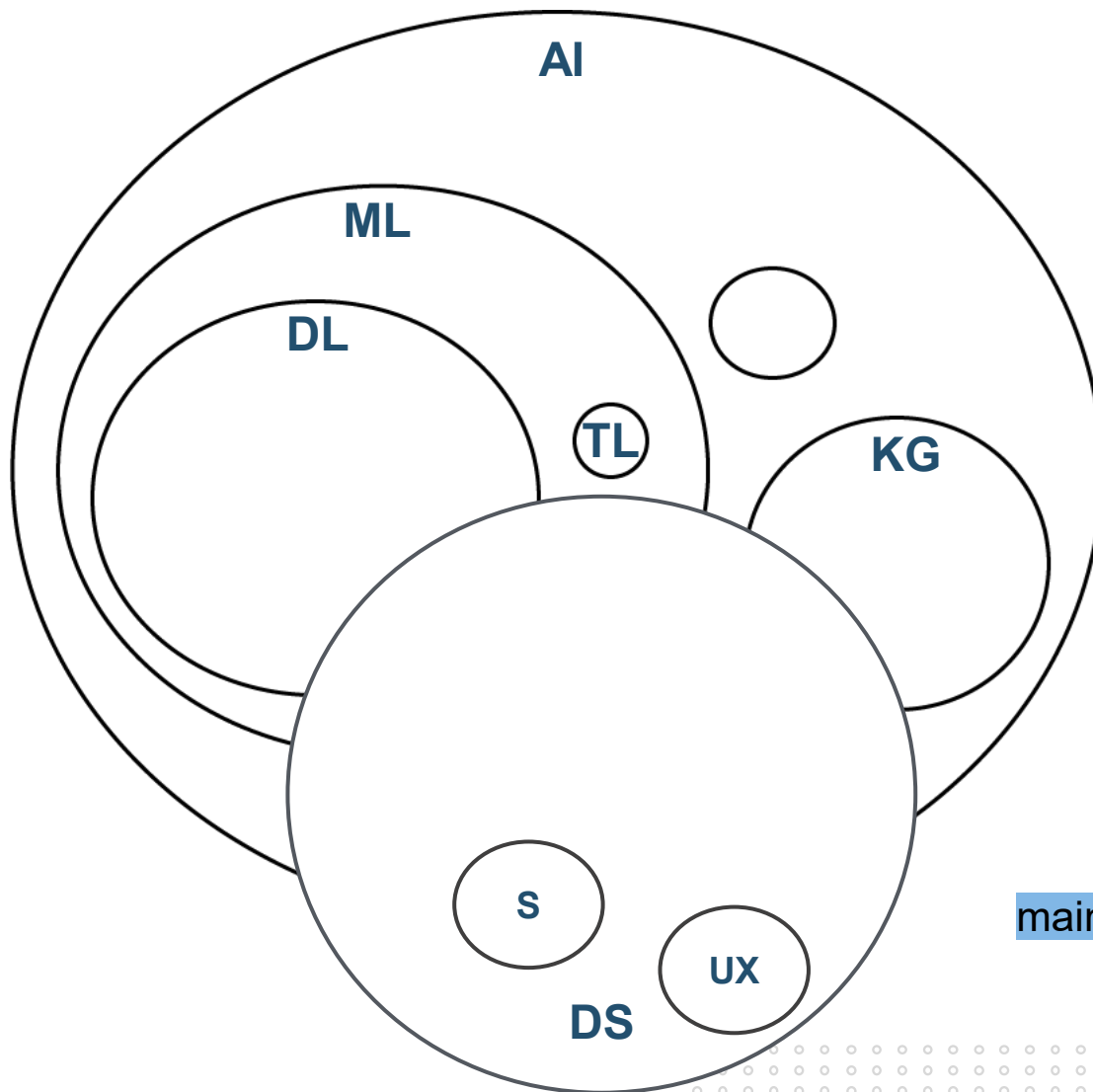
S: Statistics

UX: User eXperience

DS: Data Science

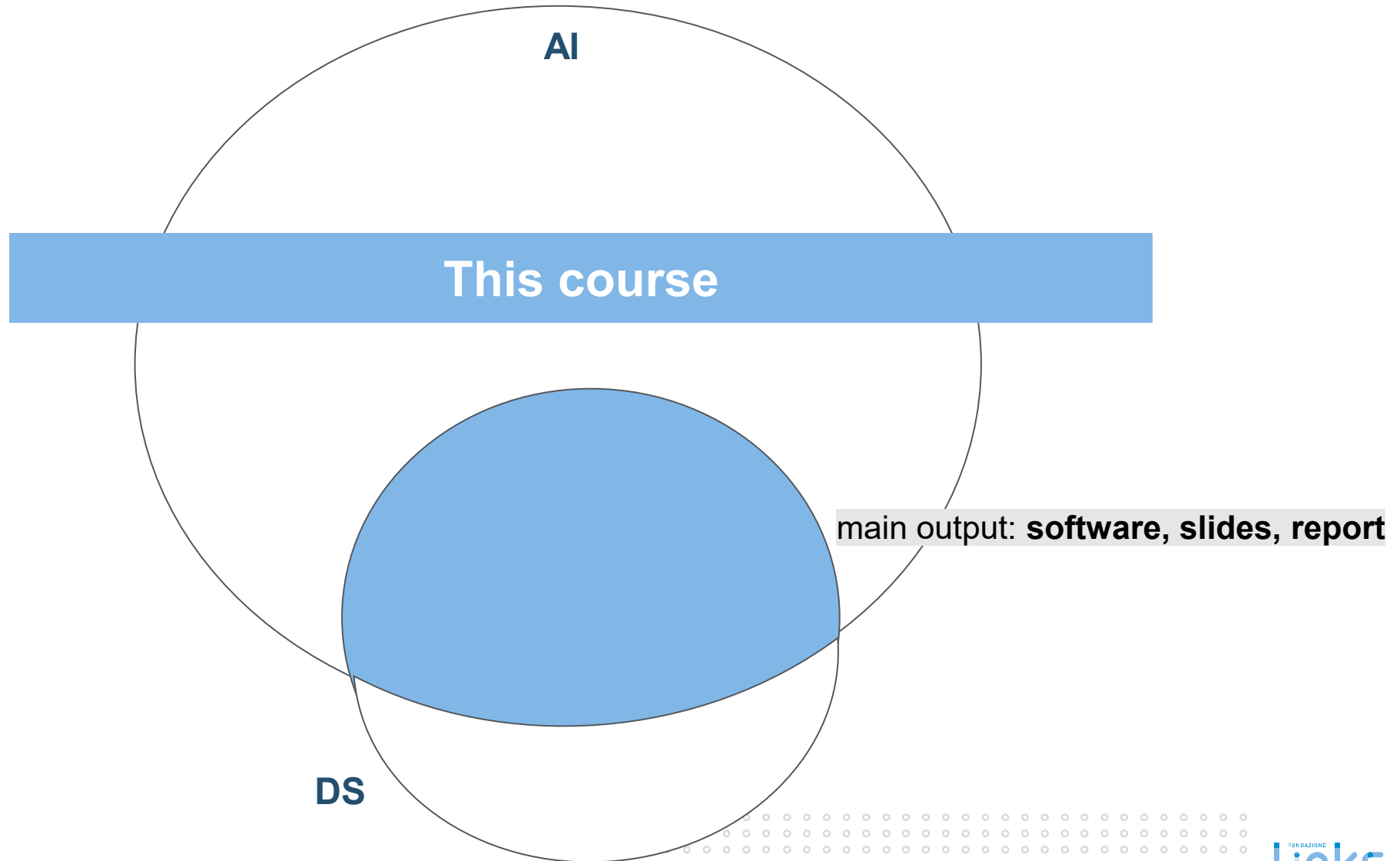
Interplay

main output:
software



main output: **visuals, slides**

Applied Data Science Project



Applied Data Science Project 101

Your goal is to develop and document an **artificial intelligence** solution by

- 1 Starting from a real-world challenge by deriving project objectives
- 2 Design a project
 - that is centered to people
 - plan the work in advance, manage the activity & monitor progress
 - select the right tools (foundation models such as a LLM, VLM)
 - customize an artificial intelligence to answer the project challenges and generate value
- 3 Quantify the benefits and impact of the project

R&I pathway

Your goal is to develop and document an **artificial intelligence** solution by

- 1 Starting from a real-world challenge and define project objectives
- 2 Design a project
 - that is centered to people
 - plan the work in advance, manage the activity & monitor progress
 - select the right tools (foundation models such as a LLM)
 - customize an artificial intelligence to answer the project challenges and generate value
- 3 Quantify the **research and innovation pathway**

Define project objectives

1

Why are you conducting the project?

Where it can be used?

Align project objectives with the SDGs



Design a project

2

Who are the users?

How a user will utilize the solution and for which task?

How the solution will look like?

Plan the work

2

“Divide and rule”

Project activities are usually grouped in:

- macro: Work Package(s)
- micro: Task(s)

with a specific due date of completion

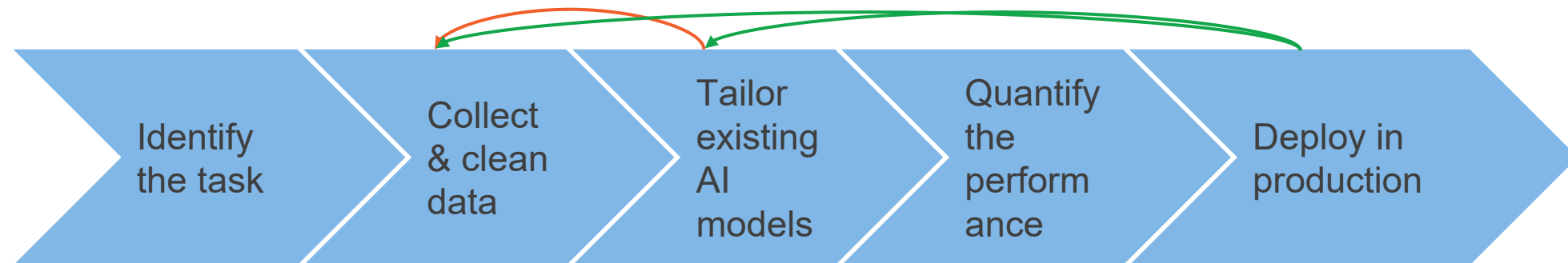
Projects have Milestones, aka dates by which it is expected a release

A project is under Risks of actions (for instance acquisition of data or recruitment of testers), the sooner the risks are identified, the clearer it is to address them

Design the solution

2

iterative processes meant to
refine the quality of the solution



From output, to outcome, to impact

3

A project generates value meaning the quantifiable benefit that the project impacts to users

The quantification of the project impact to its value is studied utilizing the concept of Key Performance Indicator(s)

Value can be framed according to economy, society, environment

Recall that a **successful project** will have the chance to apply it more on other challenges **iff**:

$$\text{value} - \text{cost of fulfilment} > 0$$

Team



Giuseppe Rizzo
Teacher



Antonella Frisiello
Teacher, Human
Computer Interaction



Alessandro Fiori
Teaching Assistant



Federico D'Asaro
Teaching Assistant



Luca Barco
Teaching Assistant

Logistics



Course topics

Lectures (26 hours)

- Building an artificial intelligence application (1.5h)
- Introduction to project pillars: management, design, development and communication (1.5h)
- Model and Data-centric projects (1.5h)
- Foundation models (1.5)
- Retrieval Augmented Generation (1.5h)
- Artificial intelligence ethics (1.5h)
- Impact of a project and SGDs (1.5h)
- Project tools (14h)
 - Management: GANTT, Work breakdown structure, work packages and tasks, milestone
 - Design: User personas, stakeholder map, functional requirements
 - Development:
 - Foundation models: Large language models and vision language models
 - Domain adaptation and downstream tasks
 - Retrieval Augmented Generation implementation
 - Version control and testing
 - Communication: Presentation, Technical Paper, Deliverable
- Success stories of past projects (1.5h)

Laboratory activities (54h)

- Project proposals (1.5h)
- User-centred application (3h)
 - Stakeholder maps and user personas
 - User journey
- Use of SDKs and REST APIs of commercial neural models with prompt engineering (6h)
- Use of Ollama with open source models (3h)
- Development of the application with tutoring (40.5)



1st week of lectures

Room	Start Time	End Time	Topic
4I	Mon Sep 22 2025 11:30:00 GMT+0200	Mon Sep 22 2025 13:00:00	L1-Introduction
ACSLab	Tue Sep 23 2025 10:00:00 GMT+0200	Tue Sep 23 2025 11:30:00	L2-Project Pillars
ACSLab	Wed Sep 24 2025 08:30:00 GMT+0200	Wed Sep 24 2025 10:00:00	L3-Models & data-centric
ACSLab	Wed Sep 24 2025 10:00:00 GMT+0200	Wed Sep 24 2025 11:30:00	L4-Foundation Models

Recommended Background

Master technologies that are linked to AI in particular **deep learning** and **machine learning**

Formal prerequisites for this course are both **software engineering** and **programming** (if you have taken one at another university, this is fine)

We strongly recommend that students have experience with **Python**, have a background in **probability** and **statistics**, and **linear algebra**

If you don't have background in these areas, please compensate in teaming up with mates who master those fields and ask for references to get up to speed (at least understand the terminology and what we are talking about)

General rule of thumb: If the project seems hard, but you have ideas about how to proceed, you probably have the right level of background; if the project seems hard and you have no idea how to proceed, this may be the wrong course (I bet this won't happen)

Course material

All course material (slides, lecture videos, project descriptions) is available on the *Portale della Didattica* of *Politecnico di Torino* course webpage:

<https://didattica.polito.it> & <https://adsp-polito.github.io>

Slides & class videos - if recorded - will be uploaded after the lecture (best effort, up to 1 day)



Additional general reading materials

- Machine Learning Yearning, by Andrew Ng
- Data Science from Scratch, Joel Grus
- Harvard Business Review Project Management Handbook: How to Launch, Lead, and Sponsor Successful Projects, by Antonio Nieto-Rodriguez
- Oxford Guide to Effective Writing and Speaking: How to Communicate Clearly, by John Seely
- The Design of Everyday Things: Revised and Expanded Edition, by Donald Norman
- Noessel C. Designing Agentive Technology. AI That Works for People. Rosenfeld, 2013

Laboratories for projects

A major part of the course is to conduct a real-world project, this accounts about $\frac{3}{4}$ of the whole course

You will be asked to express your interest as team to a list of projects.

➔ 1 team will be assigned to 1 project. A team will be of 3 students aprox

The project will be conducted under the supervision of the whole teaching team and 1 representative for each project coming for the involved companies, universities or research labs

Final report will be a latex document, slides, and a python notebook

Project delivery - WHAT

Projects are due at the end of the semester

The exact day will be communicated in the first half of the course

The delivery includes 3 outputs for each project:

- slide deck
- document
- software

Project delivery - WHERE

In a github repository of the project <https://github.com/adsp-polito> created for the project

Each repository will be public

You'll be invited to join the organization to your polito email (meaning the one that is in the portal) soon, please check your email

Part of the grading it is going to be your contribution to the repository



Project delivery - WHEN

3 intermediate checks, 15-minute presentation for each group:

- 19/11/2025
- 10/12/2025
- 07/01/2026

The final submission 1 week before the exam

Project presentation

The best way to learn a subject is to teach it

Students will prepare a presentation to be given in front of all students of the course and (possibly) the external tutors who have followed the execution of the project

Out of lecture interactions

They don't substitute lectures, nor lab activities

Drop an email, will be processed soon. However, the best way to interact is coming at lecture and ask!

If problem on projects, open a ticket through github

Grading

Team project (80%) + compulsory individual oral exam (20%)

The maximum grade for the team project is 32

The maximum grade for the oral part is 32

The final grade is given by weighted average of the two parts:

$$0.8 * \text{grade team project} + 0.2 * \text{grade individual oral part}$$

The exam is passed if the grade of the team project is greater than or equal to 18 and the grade of the oral part is greater than or equal to 18

What will be assessed?

Team project

- team project assessment is based on the performance and accuracy of the proposed solution, in terms of standard quality measures (e.g., prediction, accuracy) and completeness (i.e., in depth analysis of each phase of the designed process and motivation for selecting given techniques and algorithms). The clearness and completeness of the delivered reports will also be considered
- the grade is valid for the entire academic year

Individual oral part

- the assessment covers all the theoretical parts of the course. The score is based on the completeness and clarity of the answers
- the grade is valid for the session

Autograding

Reasoning on assessing a proper performance is the best way to be aware of the value of what has been done

Students will be asked to grade both the project teamwork (1 grade for the whole team, this means you should agree beforehand) and for your individual oral part

Along with the grade, we will expect to hear the reasons

Both grades will be considered as proposals and be considered for the final grade



Not happy with your final grade?

Project team

- you can further extend some parts that resulted be brittle of your assigned project
- in the final delivery (slides, document and software) we expect a **delta improvement** for each output. If the delta is missing, there are no conditions to have the retake
- **be aware**: support from the project mentors and from the project assistants will become best effort, thus do not pretend any help

Individual oral part

- You can retake the oral part

Grades are kept within the **academic year**

Class and lab participation

Both participations are recommended but not mandatory

We believe they are an important moment for learning and exchanging ideas and clear doubts

These moments **cannot be substituted** by questions via email or requests of meetings

Meetings

Maximize your participation during classes and labs to shed light on your doubts

External meetings (not part of the teaching duties) have an enormous costs to our teaching execution

Anyway, we offer you the chance to digging into the topics and clear doubts

A meeting is subject to our decision, and it will last no more than 30 minutes

Make sure you did your homework preparing a detailed agenda to be prepared for the meeting

Student well-being

Courses are stressful environments, we acknowledge this

In my experience, most student difficulties are the product of stress and last-minute decisions
Is there a need to get to that point?

Don't sacrifice quality of life for this course, which remains a course where **to learn** and not to suffer. Enjoy this moment

Sleep, eat well, exercise, socialize and have fun when working at the project, these are key for success



Thank you for your attention.

Questions?



CONTACTS

Giuseppe Rizzo

Adjunct Professor (Politecnico di Torino)
Program Manager (LINKS Foundation)

giuseppe.rizzo@polito.it

FONDAZIONE LINKS
Via Pier Carlo Boggio 61 | 10138 Torino
P. +39 011 22 76 150
LINKSFOUNDATION.COM